

20/12/22

Computer Fundamentals

Wireless process $\rightarrow$ Antenna.

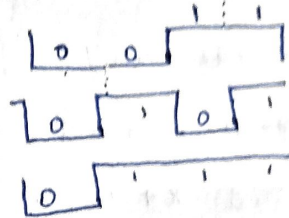
Frequency range of bluetooth, radio, satellite TV, wifi, remote, LAN, Remote.

Logic Gates (symbolic representation)

1) OR



X	0	0	1	1
Y	0	1	0	1
F	0	1	1	1



2) AND



X	0	0	1	1
Y	0	1	0	1
F	0	0	0	1

3) NOT



X	0	1
F	1	0

4) (Buffer)

X	0	1
F	0	1

5) NAND



X	0	0	1	1
Y	0	1	0	1
F	1	1	0	0

NAND - even single input is 0
o/p is 1

6) NOR



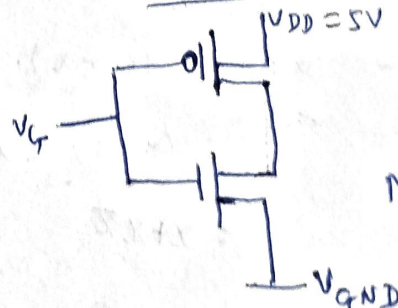
X	0	0	1	1
Y	0	1	0	1
F	1	0	0	0

Analog - sinusoidal
Digital - Pulses.

7) EX-OR $F = X \oplus Y$

8) EX-NOR $F = X \odot Y$

CMOS $\rightarrow$ Application of Transistor



NOT.

V_{in}	V_o
X	F
0	1
1	0

Boolean Algebra Postulates

Postulate 2 (a) $x + 0 = x$ (b) $x \cdot 1 = x$

5 (a) $x + \bar{x} = 1$ (b) $x \bar{x} = 0$

3 (a) $x + y = y + x$ (b) $x \cdot y = y \cdot x$

commutative law

Postulate 4 (a) $x(y + z) = xy + xz$

distribution law (b) $x + yz = (x + y)(x + z)$

theorem 1 (a) $x + x = x$ (b) $x \cdot x = x$

2 (a) $x + 1 = 1$ (b) $x \cdot 0 = 0$

3 (a) $\overline{\bar{x}} = x$

4 (a) $x + (y + z) = (x + y) + z$

(b) $x(yz) = (xy)z$

5 (a) $\overline{x + y} = \bar{x} \cdot \bar{y}$

(b) $\overline{xy} = \bar{x} + \bar{y}$

6. (a) $x + xy = x \rightarrow x \cdot 1 + xy \rightarrow x(1 + y) = x \cdot 1$

(b) $x(x + y) = x$

Thm 1 (a) $x + x = x$

LHS:- $x + x = (x + x) \cdot 1$ ($\because x \cdot 1 = x$)

~~x~~ $= (x + x) \cdot (x + \bar{x})$ ($\because x + \bar{x} = 1$)

$= x + x\bar{x}$ ($\because x + yz = (x + y)(x + z)$)

$= x + 0$ ($\because x\bar{x} = 0$)

$= x$ ($\because x + 0 = x$)

$= \text{RHS}$

x	$\bar{x}$	$x + \bar{x}$
0	1	1
1	0	1

Thm 1(b) $x \cdot x = x$

LHS $x \cdot x$

$x \cdot x + 0$ ($\because x + 0 = x$)

$x \cdot x + x \cdot \bar{x}$ ($\because x \bar{x} = 0$)

$x(x + \bar{x})$ ($\because x(y + z) = xy + xz$)

$x \cdot 1$ ($\because x + \bar{x} = 1$)

$x = \text{RHS}$ ($\because x \cdot 1 = x$)

Thm 2 (a) $x + \bar{x} = 1$

LHS $x + \bar{x}$

$x + (x + \bar{x})$ ($\because x + \bar{x} = 1$)

~~$x + x + \bar{x}$~~

~~$= (x + x) \cdot 1 + \bar{x}$ ($\because (x + x) = (x + x) \cdot 1$)~~

~~$= (x + x)$~~

$x + (1 \cdot \bar{x})$ ($\because x + yz = (x + y)(x + z)$)

$x + \bar{x}$ ($\because x \cdot 1 = x$)

$1 = \text{RHS}$ ($\because x + \bar{x} = 1$)

Thm 2(b) $x \cdot 0 = 0$ (duality principle)

LHS:-

~~$x \cdot x$~~
 $(x \cdot 0) + 0$ ($\because x + 0 = x$)

$(x \cdot 0) + (x \bar{x})$ ($\because x \bar{x} = 0$)

$x + (0 \cdot \bar{x})$ ($\because x + yz = (x + y)(x + z)$)

Thm 3(a)

Thm 4(a)

x	y	z	$(x+y)$	$(y+z)$	$x+(y+z)$	$(x+y)+z$
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	0	1	1
1	0	0	1	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Thm 6 (a)

$$\overline{x+y} = \bar{x} \cdot \bar{y}$$

T.T.

x	y	x+y	$\overline{x+y}$	$\bar{x}$	$\bar{y}$	$\bar{x} \cdot \bar{y}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

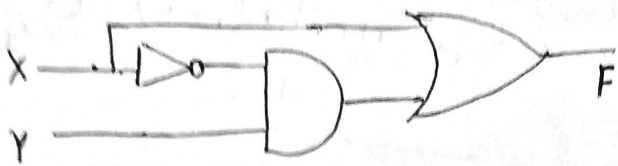
Combinational Circuit (Thm cannot be used).

$$x + \bar{x}y$$

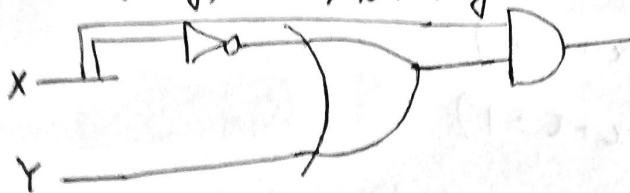
simplification

$$(x + \bar{x})(x + y) = 1 \cdot (x + y)$$

$$(\because x + yz = (x + y)(x + z))$$



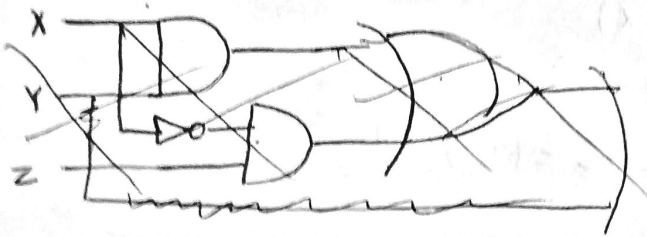
$$x(\bar{x} + y) = x\bar{x} + xy$$



$$F = x \cdot y$$



$$xy + \bar{x}z + yz$$



$$xy + \bar{x}z + yz$$

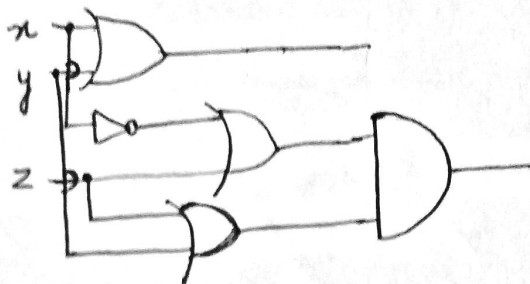
$$xy + \bar{x}z + (x + \bar{x})yz$$

$$xy + \bar{x}z + xyz + \bar{x}yz$$

$$xy(1 + z) + \bar{x}z(1 + y)$$

$$xy + \bar{x}z \cdot 1 + (4 \text{ gates})$$

$$(x+y)(\bar{x}+z)(y+z)$$



h

17/1/2023

Computer Fundamentals - class

$$F_1 = XYZ$$

$$F_2 = X + \bar{Y}Z$$

$$F_3 = \bar{X}\bar{Y}Z + \bar{X}YZ + X\bar{Y}$$

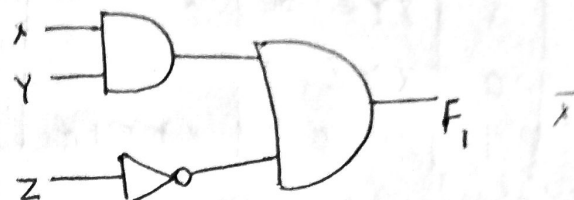
001 011 100

$$F_4 = X\bar{Y} + \bar{X}Z$$

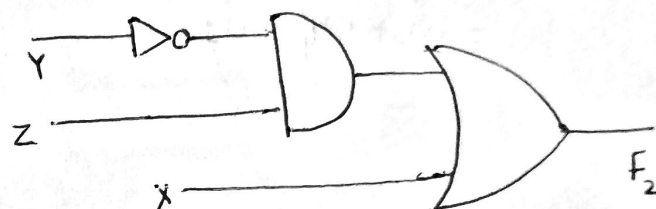
0
1
110
Two AND Two OR NOT

x	y	z	F ₁	F ₂	F ₃	F ₄
0	0	0	0	0	0=0+0+0	0
0	0	1	0	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	0

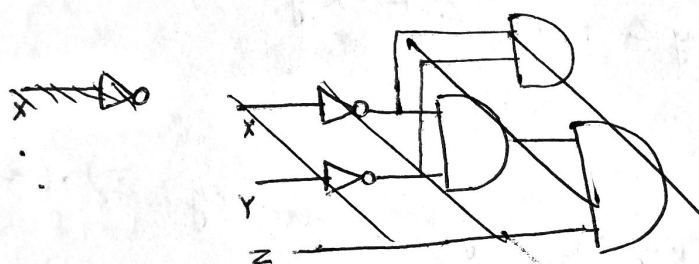
(a)



(b)

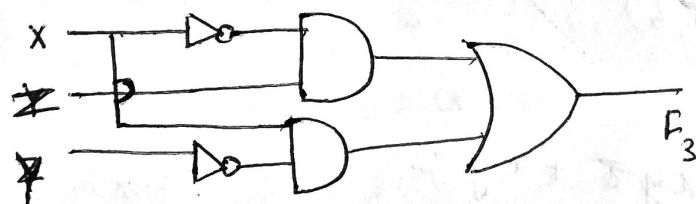


(c)



$$\bar{X}Z(\bar{Y}Z + Y) + X\bar{Y}$$

$$\bar{X}Z + X\bar{Y}$$



(d) $F_4 = X\bar{Y} + \bar{X}Z$ (same as F_3)

Maxterms & Minterms

x	y	minterm	maxterm
0	0	$\bar{x}\bar{y}$ m_0	$x+y$ M_0
0	1	$\bar{x}y$ m_1	$x+\bar{y}$ M_1
1	0	$x\bar{y}$ m_2	$\bar{x}+y$ M_2
1	1	xy m_3	$\bar{x}+\bar{y}$ M_3

$$F_1 = x \cdot y + \bar{x}y = m_3 + m_1$$

$$= \text{S.O.P}$$

$$= \sum(m_1, m_3)$$

$$F = (x+y)(\bar{x}+\bar{y}) = M_0 \cdot M_2$$

$$= \pi(M_0, M_2)$$

X	Y	Z	Min term	Max term		F ₁	F ₂
0	0	0	$\bar{x}\bar{y}\bar{z}$ m ₀	$\bar{x}+\bar{y}+\bar{z}$ M ₀		0	0
0	0	1	$\bar{x}\bar{y}z$ m ₁	$\bar{x}+\bar{y}+z$ M ₁		1	0
0	1	0	$\bar{x}y\bar{z}$ m ₂	$\bar{x}+y+\bar{z}$ M ₂		0	0
0	1	1	$\bar{x}yz$ m ₃	$\bar{x}+y+z$ M ₃		0	1
1	0	0	$x\bar{y}\bar{z}$ m ₄	$x+\bar{y}+\bar{z}$ M ₄		1	0
1	0	1	$x\bar{y}z$ m ₅	$x+\bar{y}+z$ M ₅		0	1
1	1	0	$xy\bar{z}$ m ₆	$x+y+\bar{z}$ M ₆		0	1
1	1	1	xyz m ₇	$x+y+z$ M ₇		1	1

1+3+0
9+1+4
2 p.w.

$$F_1 = \bar{x}\bar{y}z + x\bar{y}\bar{z} + xyz$$

$$= m_1 + m_4 + m_7$$

$$= \Sigma(1, 4, 7)$$

~~Max term~~

$$\bar{F}_1 = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

$$= m_0 + m_2 + m_3 + m_5 + m_6$$

Max term

$$\bar{F}_1 = (x+y+z) \cdot (x+\bar{y}+z) \cdot (x+\bar{y}+\bar{z}) \cdot (\bar{x}+y+\bar{z}) \cdot (\bar{x}+\bar{y}+z)$$

$$= \Pi(0, 2, 3, 5, 6)$$

$$F_2 = \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$$

$$= m_3 + m_5 + m_6 + m_7$$

$$= \Sigma(3, 5, 6, 7)$$

$$\bar{F}_2 = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z}$$

$$= m_0 + m_1 + m_2 + m_4$$

$$= \Sigma(0, 1, 2, 4)$$

(max term)

$$\bar{F}_2 = (x+y+z) \cdot (x+\bar{y}+z) \cdot (x+\bar{y}+\bar{z}) \cdot (\bar{x}+y+z)$$

$$= \Pi(0, 1, 2, 4)$$

$$\textcircled{1} F = A + \bar{B}C$$

$$= A \cdot (B + \bar{B}) (C + \bar{C}) + (A + \bar{A}) \bar{B}C$$

$$= \underset{111}{ABC} + \underset{110}{AB\bar{C}} + \underset{101}{A\bar{B}C} + \underset{100}{A\bar{B}\bar{C}} + \underset{001}{\bar{A}\bar{B}C} + \underset{000}{\bar{A}\bar{B}\bar{C}}$$

$$= m_7 + m_6 + m_5 + m_4 + m_1 + m_0$$

$$= \Sigma(0, 1, 4, 5, 6, 7)$$

$$\bar{F} = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

$$= m_0 + m_1 + m_2 + m_3 + m_4 + m_5$$

$$F = A + \bar{B}C \quad (\text{w/o complement find max term})$$

$$= (A + \bar{B})(A + C)$$

$$= (A + \bar{B} + C\bar{C})(A + B\bar{B} + C) \quad (\text{since } x\bar{x} = 0)$$

$$= (A + \bar{B} + C)(A + B + C)(A + \bar{B} + \bar{C})$$

$$= M_2 \cdot M_3 \cdot M_0 \cdot M_3$$

$$= \Pi(M_0, M_2, M_3)$$

$$\textcircled{2} F = xy + \bar{x}z$$

$$= xy(z + \bar{z}) + \bar{x}(y + \bar{y})z$$

$$= xyz + x\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}z$$

$$= m_7 + m_6 + m_3 + m_1$$

$$= \Sigma(1, 3, 6, 7)$$

$$F = (xy + \bar{x})(xy + z)$$

$$= (\bar{x} + x)(\bar{x} + y)(x + z)(y + z)$$

$$= (\bar{x} + y)(x + z)(y + z)$$

$$= (\bar{x} + y + z\bar{z})(x + y\bar{y} + z)(x\bar{x} + y + z)$$

$$= (\bar{x} + y + z)(\bar{x} + y + \bar{z})(x + y + z)(x + \bar{y} + z)(x + y + \bar{z})(\bar{x} + y + z)$$

$$= M_4 \cdot M_5 \cdot M_0 \cdot M_2 = \Pi(M_0, M_2, M_4, M_5)$$

25/1/23

x	y			f_1	f_2	f_3
0	0	m_0	$\bar{x}\bar{y}$	1	1	0
0	1	m_1	$\bar{x}y$	1	0	1
1	0	m_2	$x\bar{y}$	0	1	0
1	1	m_3	xy	0	0	1

x	y	
0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3

x	$\bar{y}$	y
$\bar{x}$	$\bar{x}\bar{y}$	$\bar{x}y$
x	$x\bar{y}$	xy

x	y	
0	0	1
0	1	1
1	0	0
1	1	0

$$f_1 = \bar{x}\bar{y} + \bar{x}y$$

$$= \bar{x}(\bar{y} + y)$$

$$= \bar{x} \cdot 1$$

$$= \bar{x}$$

$$f_2 = \bar{x}\bar{y} + x\bar{y}$$

$$= \bar{y}(\bar{x} + x)$$

$$= \bar{y} \cdot 1$$

$$= \bar{y}$$

($\because \bar{x} + x = 1$)

($\because x \cdot 1 = x$)

x	y	
0	0	1
0	1	1
1	0	0
1	1	0

$$f_3 = \bar{x}y + xy$$

$$= y(\bar{x} + x)$$

$$= y \cdot 1$$

$$= y$$

($\because x + \bar{x} = 1$)

($\because y \cdot 1 = y$)

x	y	
0	0	0
0	1	1
1	0	0
1	1	1

* one variable will be common for two blocks.

x	y	z		f_1	f_2	f_3
0	0	0	$\bar{x}\bar{y}\bar{z}$ m_0	1	1	1
0	0	1	$\bar{x}\bar{y}z$ m_1	1	1	0
0	1	0	$\bar{x}y\bar{z}$ m_2	0	0	1
0	1	1	$\bar{x}yz$ m_3	0	0	0
1	0	0	$x\bar{y}\bar{z}$ m_4	0	1	1
1	0	1	$x\bar{y}z$ m_5	0	0	0
1	1	0	$xy\bar{z}$ m_6	0	0	1
1	1	1	xyz m_7	0	0	0

yz	00	01	11	10
x	$\bar{x}\bar{y}\bar{z}$ m_0	$\bar{x}\bar{y}z$ m_1	$\bar{x}y\bar{z}$ m_2	$\bar{x}yz$ m_3
0	1	1	0	0
1	$x\bar{y}\bar{z}$ m_4	$x\bar{y}z$ m_5	$xy\bar{z}$ m_6	xyz m_7
1	0	0	0	0

$\bar{x}\bar{y}$

$$\begin{aligned}
 F_1 &= \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z \\
 &= \bar{x}\bar{y}(\bar{z}+z) \\
 &= (\bar{x}\bar{y}) \cdot 1 \\
 &= \bar{x}\bar{y}
 \end{aligned}$$

$$\begin{aligned}
 F_2 &= \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} \\
 &= \bar{x}\bar{y}(\bar{z}+z) + x\bar{y}\bar{z} \\
 &= \bar{x}\bar{y}(1) + x\bar{y}\bar{z} \\
 &= \bar{y}(\bar{x} + x\bar{z})
 \end{aligned}$$

x \ yz	00	01	11	10
0	1	1	0	0
1	1	0	0	0

$$\bar{y}(\bar{x}\bar{z} + \bar{x}z + x\bar{z})$$

$$\bar{y}(\bar{x}(\bar{z}+z) + x\bar{z})$$

$$\bar{y}(\bar{x} + x\bar{z})$$

$$\bar{y}(\bar{x}+x)(\bar{x}+\bar{z})$$

$$\bar{y}(\bar{x}+\bar{z})$$

$$\bar{x}\bar{y} + \bar{z}\bar{y}$$

$$F_2 = \bar{x}\bar{y} + \bar{y}\bar{z}$$

$$\begin{aligned}
 F_3 &= \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x y \bar{z} \\
 &= \bar{x}\bar{z}(\bar{y}+y) + x\bar{z}(\bar{y}+y) \\
 &= \bar{x}\bar{z} + x\bar{z} \\
 &= \bar{z}(\bar{x}+x) \\
 &= \bar{z}
 \end{aligned}$$

x \ yz	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$F = \bar{x}y\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$$

x \ yz	00	01	11	10
0	0	0	1	1
1	1	1	0	0

$$= \bar{x}\bar{y} + \bar{x}y = x \oplus y$$

$$\bar{x}\bar{z}(\bar{y}+y)$$

$$= \bar{x}\bar{y}(\bar{z}+z) + x\bar{y}(\bar{z}+z)$$

$$= \bar{x}\bar{y} + x\bar{y}$$

$$= x \oplus y$$

* all the 1's should be considered atleast once.

$$F_2 = \bar{x}yz + x\bar{y}\bar{z} + xyz + xy\bar{z}$$

$$= \bar{x}yz + x\bar{y}\bar{z} + xy(z + \bar{z})$$

$$= \bar{x}yz + x\bar{y}\bar{z} + xy$$

31/1/23

	w	x	y	z
wx \ yz	00	01	11	10
00	m ₀	m ₁	m ₃	m ₂
01	m ₄	m ₅	m ₇	m ₆
11	m ₁₂	m ₁₃	m ₁₅	m ₁₄
10	m ₈	m ₉	m ₁₁	m ₁₀

$$F(w, x, y, z) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 13, 14)$$

wx \ yz	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$	1	1	0	1
$\bar{w}x$	1	1	0	1
$w\bar{x}$	1	1	0	1
wx	1	1	0	0

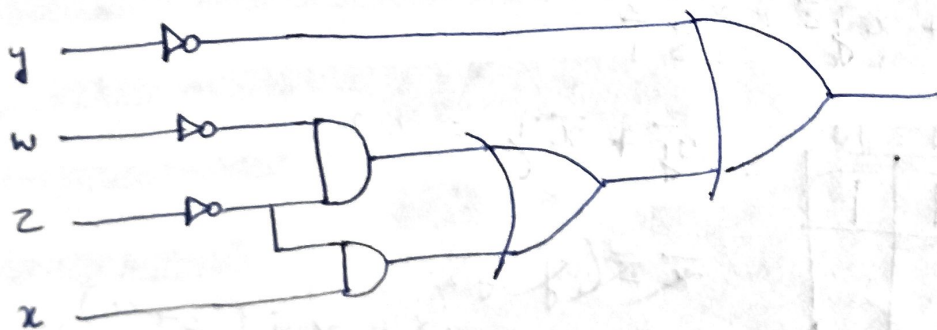
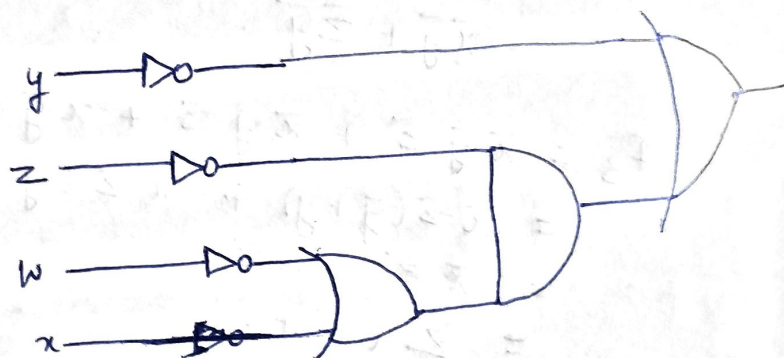
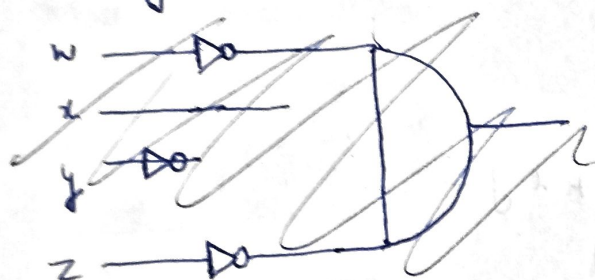
$$F = \bar{y} + \bar{w}\bar{z} + x\bar{z}$$

$$= \bar{y} + \bar{z}(\bar{w} + x)$$

$$\bar{y} + y\bar{z} + \cancel{xy\bar{z}} \quad y\bar{z} = \bar{y} + y\bar{z}$$

$$\bar{y} + y\bar{z}(1 + \cancel{wx})$$

$$\bar{y} + y\bar{z}$$



$$11C_2$$

$$\frac{11 \times 105}{2 \times 2} = 115$$

* Calculate gates when minimization is not done?

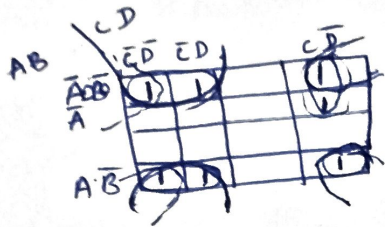
$$F = \bar{A}\bar{B}C + \bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C$$

Minterms.

$$F = \bar{A}\bar{B}C(D+\bar{D}) + (A+\bar{A})\bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C(D+\bar{D})$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + A\bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}CD + A\bar{B}C\bar{D}$$

$\begin{matrix} 10001 & 00000 & 10100 & 00100 & 01100 & 10001 & 10000 \\ & 0 & 10 & 2 & 6 & 7 & 8 \end{matrix}$



$$\bar{A}\bar{B}C + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C\bar{D}$$

$$\bar{B}C + \bar{B}\bar{D} + \bar{A}C\bar{D}$$

Odd function — No. of 1 must be odd in f^n .

x	y	F
0	0	0
0	1	1
1	0	1
1	1	0

x	y	$\bar{y}$
0	0	1
1	0	1
0	1	0
1	1	0

$$x\bar{y} + \bar{x}y = x \oplus y$$

x	y	z	F	
0	0	0	0	m_0
0	0	1	1	m_1
0	1	0	1	m_2
0	1	1	0	m_3
1	0	0	1	m_4
1	0	1	0	m_5
1	1	0	0	m_6
1	1	1	1	m_7

	$y\bar{z}$	$\bar{y}z$	$y\bar{z}$	$\bar{y}z$
$\bar{x}$		1		1
x	1		1	
	0	1	3	2
	4	5	7	6

$$F = x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}z$$

$$= \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$$

$$= \bar{x}(\bar{y}z + y\bar{z}) + x(\bar{y}\bar{z} + \bar{y}z)$$

$$= \bar{x}(y \oplus z) + x(\overline{y \oplus z})$$

$$= \bar{x}(y \oplus z) + x(\overline{y \oplus z})$$

$$= x \oplus (y \oplus z)$$

$$\overline{y \oplus z} = \overline{y\bar{z} + \bar{y}z}$$

$$= (\overline{y\bar{z}})(\overline{\bar{y}z})$$

$$= (\bar{y} + z)(y + \bar{z})$$

$$= 0 + \bar{y}z + y\bar{z} + 0$$

$$= \bar{y}z + y\bar{z}$$

$$F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$$

$$d(w, x, y, z) = \Sigma(0, 2, 5)$$

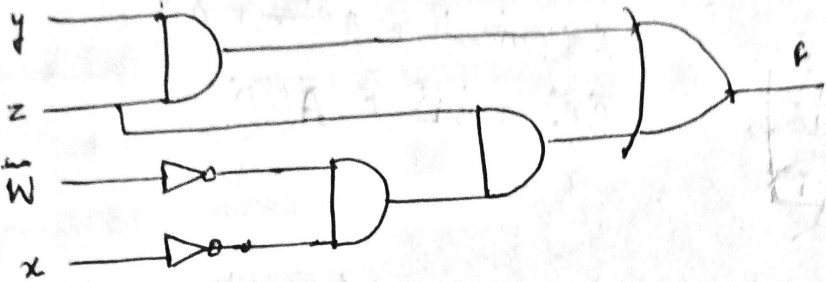
don't care terms

wx \ yz	00	01	11	10
00	X	1	1	X
01	0	X	1	0
11	0	0	1	0
10	0	0	1	0

$$F_d = \bar{w}\bar{x} + yz \quad (\text{with don't care})$$

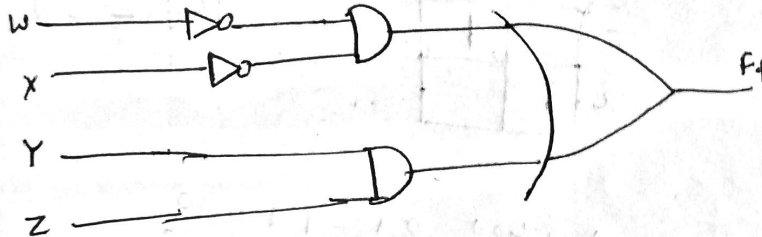
$$F = yz + \bar{w}\bar{x}$$

F:



6 gates

F_d



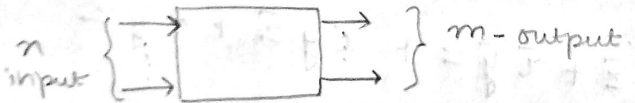
5 gates

13/02/23

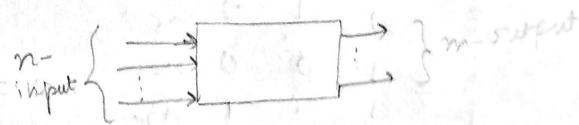
Combinational & sequential circuits

Assignment: solve of
Molam at show and
arrange them

Combinational circuits.



Sequential circuit

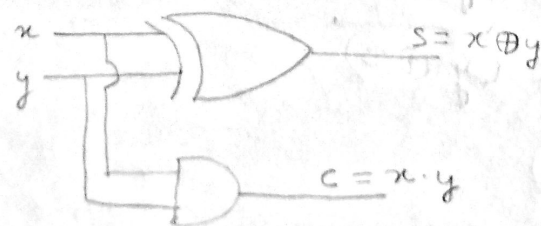


Half adder

x	y	s	c
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$s = x\bar{y} + \bar{x}y = x \oplus y$$

$$c = x \cdot y$$



Full Adder

x	y	z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

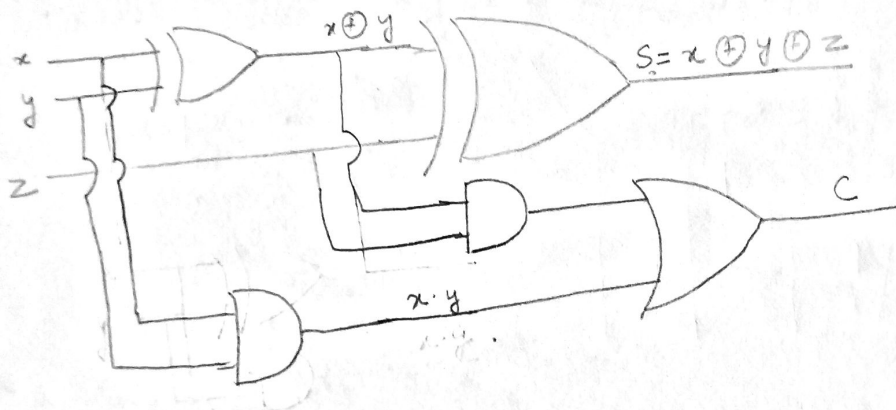
		1		1
0	1		1	
1		1		

$$\begin{aligned}
 &= \bar{x}y\bar{z} + \bar{x}y z + x\bar{y}\bar{z} + x\bar{y} z + x y \bar{z} + x y z \\
 &= \bar{x}y z + \bar{x}y \bar{z} + x\bar{y} z + x y z \\
 &= \bar{x}(\bar{y} + y)z \\
 &= \bar{x}(1)z = \bar{x}z \\
 &= \bar{x}(y \oplus z) + 1(\bar{y} + y)z + xy \\
 &= \bar{x}(y \oplus z) + (\bar{y} + y)z \\
 &S = \bar{x}(y \oplus z) + xz \quad S = x \oplus y \oplus z
 \end{aligned}$$

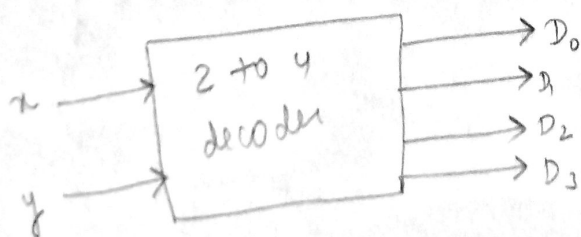
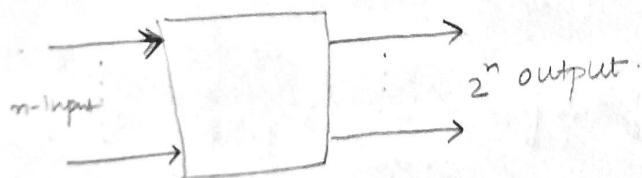
	$\bar{y}z$	$y\bar{z}$	yz	$y\bar{z}$
$\bar{x}$	0	1	1	0
x	1	1	1	0

$$C = yz + xz + xy$$

$$\begin{aligned}
 C &= \bar{x}yz + x\bar{y}z + x y \bar{z} + x y z \\
 &= z(\bar{x}y + x\bar{y}) + xy(\bar{z} + z) \\
 &= z(x \oplus y) + xy
 \end{aligned}$$

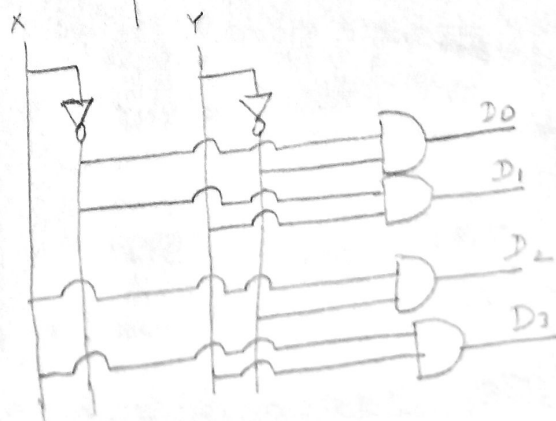


Decoder



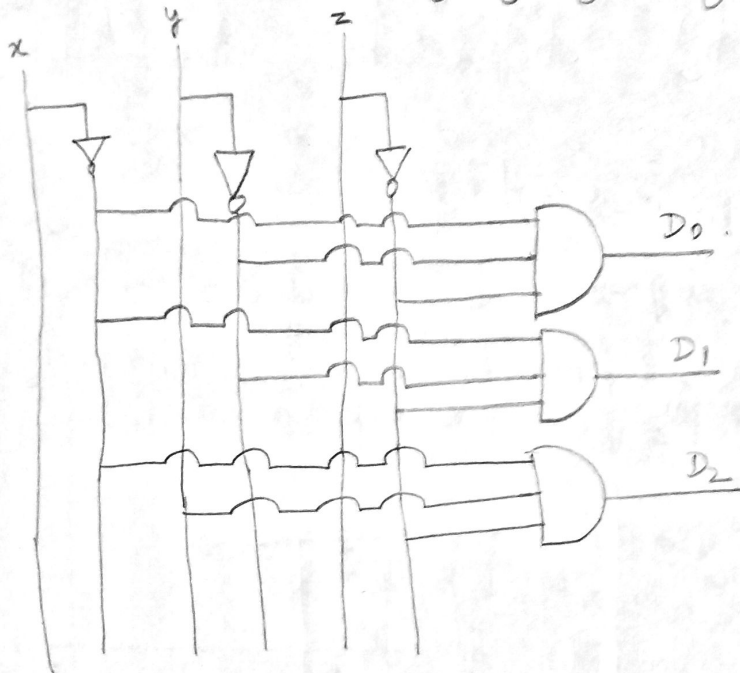
2-input

x	y	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



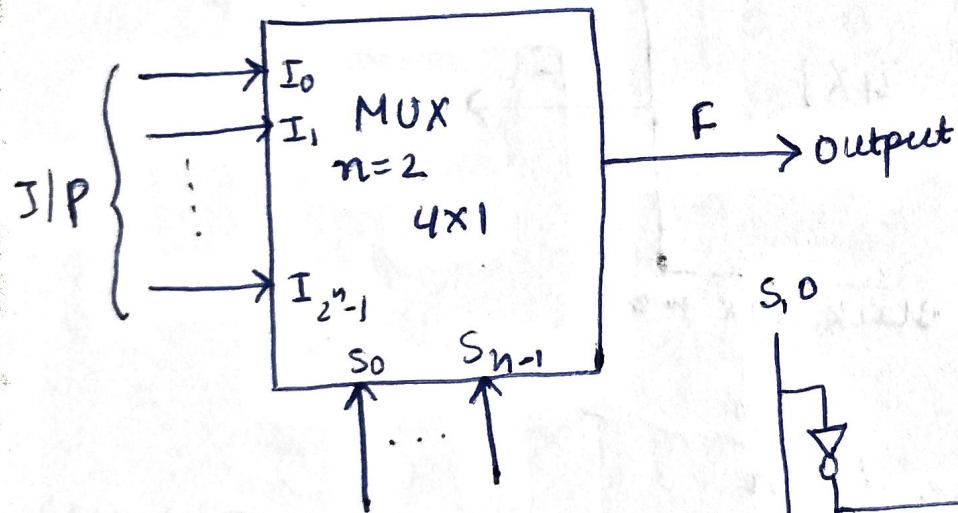
3-input decoder

x	y	z	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇	D ₈
0	0	0	1	0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	0



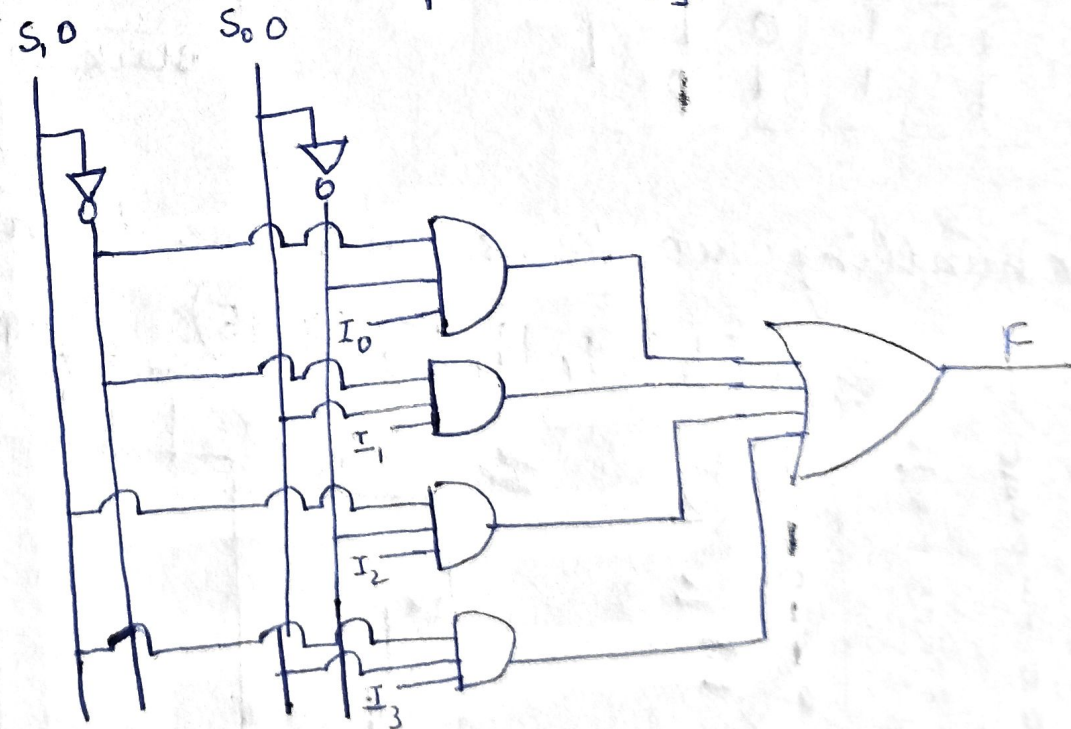
14/02/23

MUX

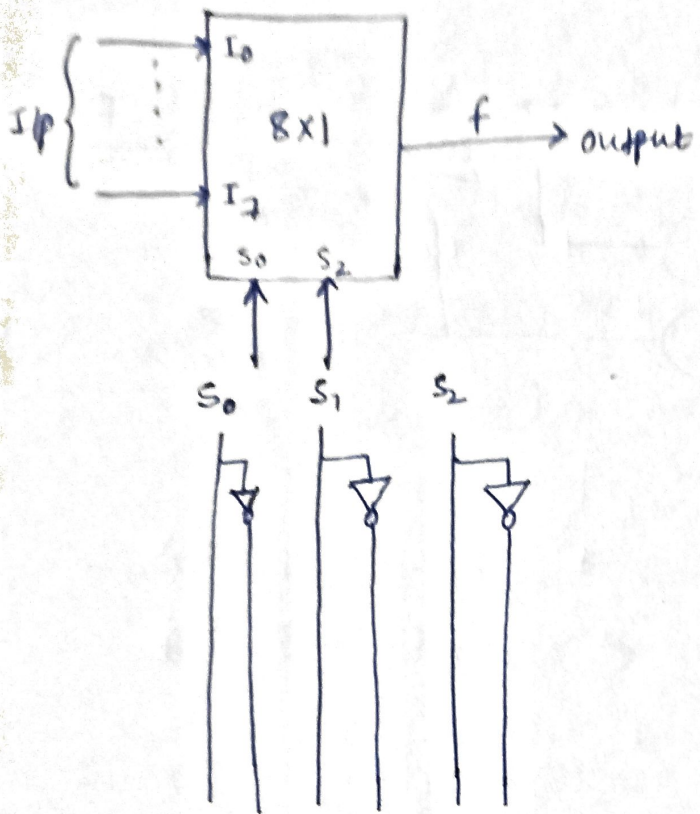


S_0	S_1	F
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

{ Mid-sem }
- 2
:- Next week
24th Feb }



8X1 MUX $n=3$

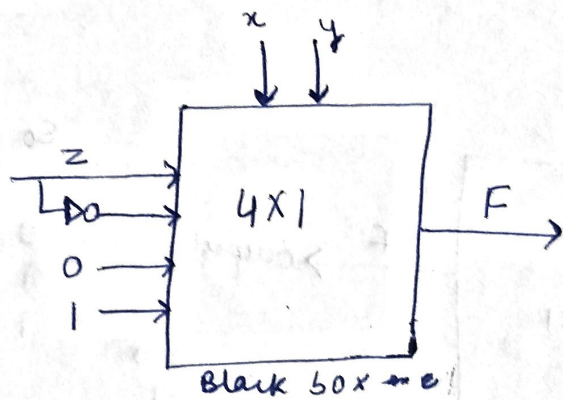


S_0	S_1	S_2	F
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

11 to 12 u

Q - $F(x, y, z) = \sum m(1, 2, 6, 7)$. Implement F using 4X1 MUX.

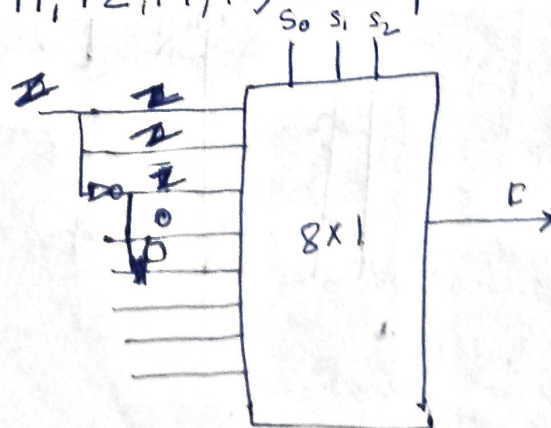
x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



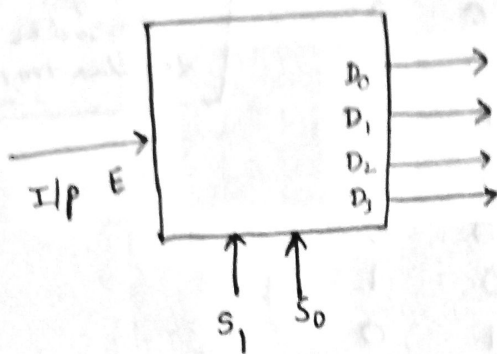
* handling two terms with one term.

Q - $F = \sum m(1, 3, 4, 11, 12, 14, 15)$. Implement F using 8X1 MUX.

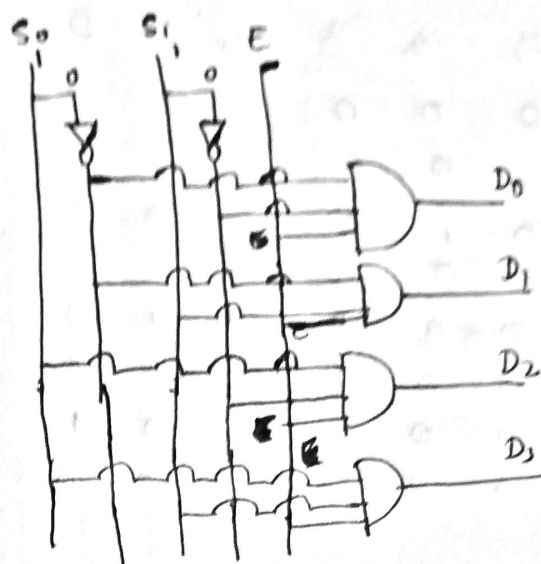
w	x	y	z	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0



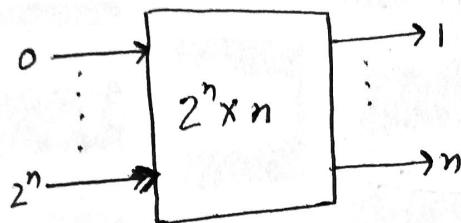
DeMux (Demultiplexer)



S_0	S_1	O/p
0	0	E
0	1	E
1	0	E
1	1	E



21/2/23

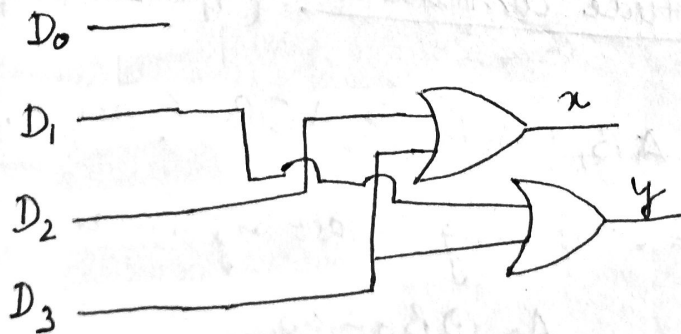


Encoder

D_0	D_1	D_2	D_3	x	y
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

$$x = D_2 + D_3$$

$$y = D_1 + D_3$$



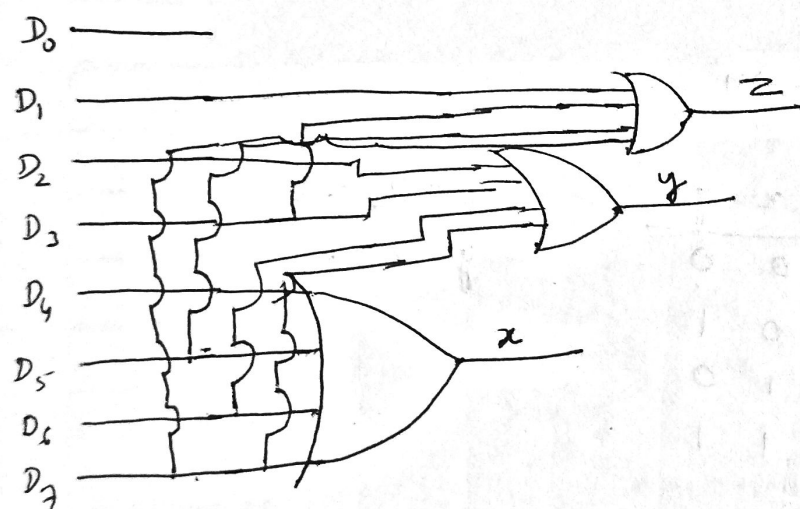
Assignment
Priority
encoder
deadline tomorrow

	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
0	1	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	1
2	0	0	1	0	0	0	0	0	0	1	0
3	0	0	0	1	0	0	0	0	0	1	1
4	0	0	0	0	1	0	0	0	1	0	0
5	0	0	0	0	0	1	0	0	1	0	1
6	0	0	0	0	0	0	1	0	1	1	0
7	0	0	0	0	0	0	0	1	1	1	1

$$x = D_4 + D_5 + D_6 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$z = D_1 + D_3 + D_5 + D_7$$



Assignment

Magnitude Comparator (Question on final exam)

A	B	
0	0	$A=B$
0	1	$A < B$
1	0	$A > B$
1	1	$A=B$

A_1	A_0	B_1	B_0
1	0	0	1
0	1	0	0

$$A_1 B_1 + \bar{A}_1 \bar{B}_1 \Rightarrow A \odot B \text{ (they are equal)}$$

* Comparing most significant bits

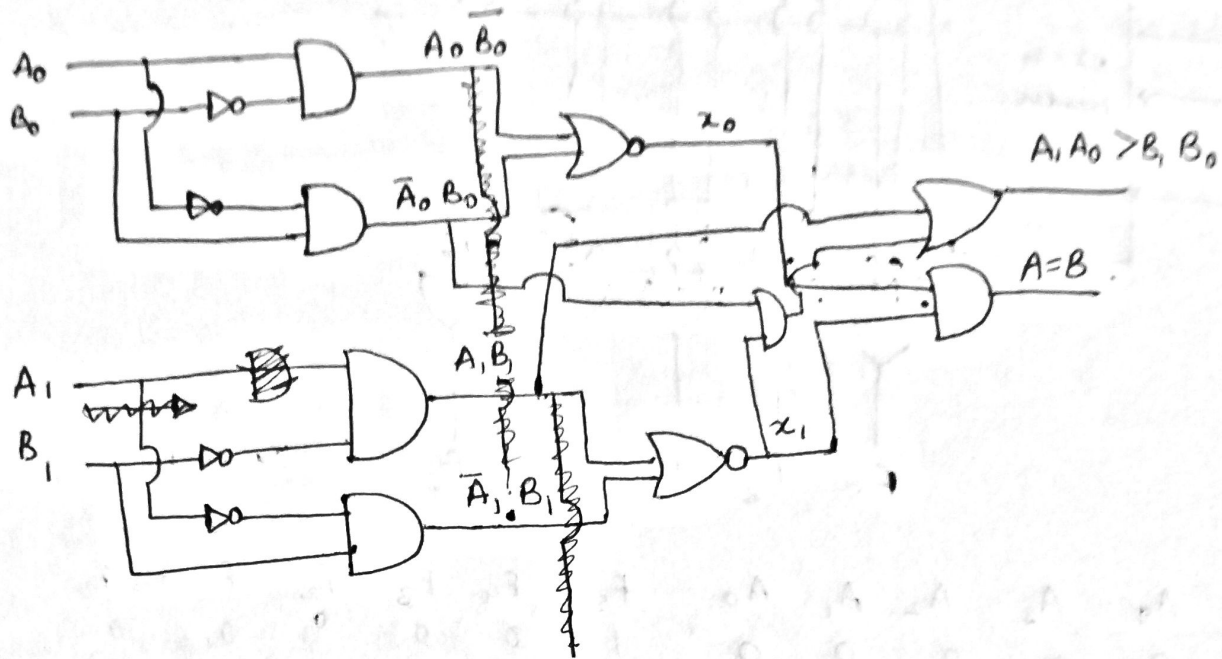
$$A_0 \odot B_0 = x_0$$

$$A_1 \odot B_1 = x_1$$

$$A_1 \bar{B}_1 + x_1 A_0 \bar{B}_0 \quad \left\{ \text{Here } A_1 A_0 > B_1 B_0 \right.$$

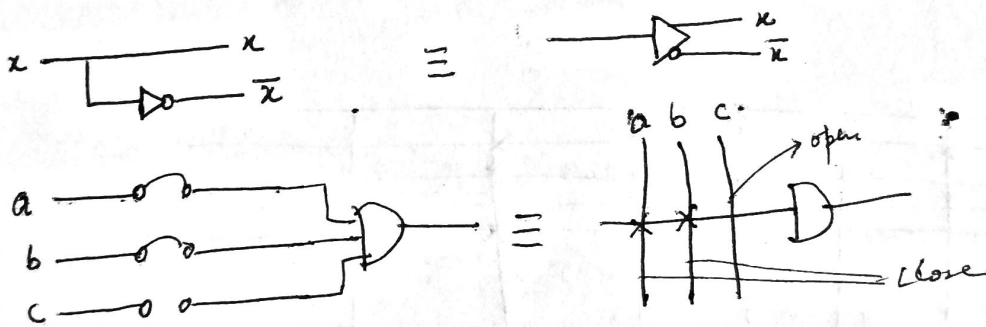
$$\bar{A}_1 B_1 + x_1 \bar{A}_0 B_0 \quad \left\{ \text{Here } A_1 A_0 < B_1 B_0 \right.$$

Implementation of Comparators

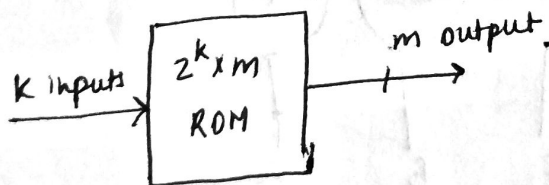


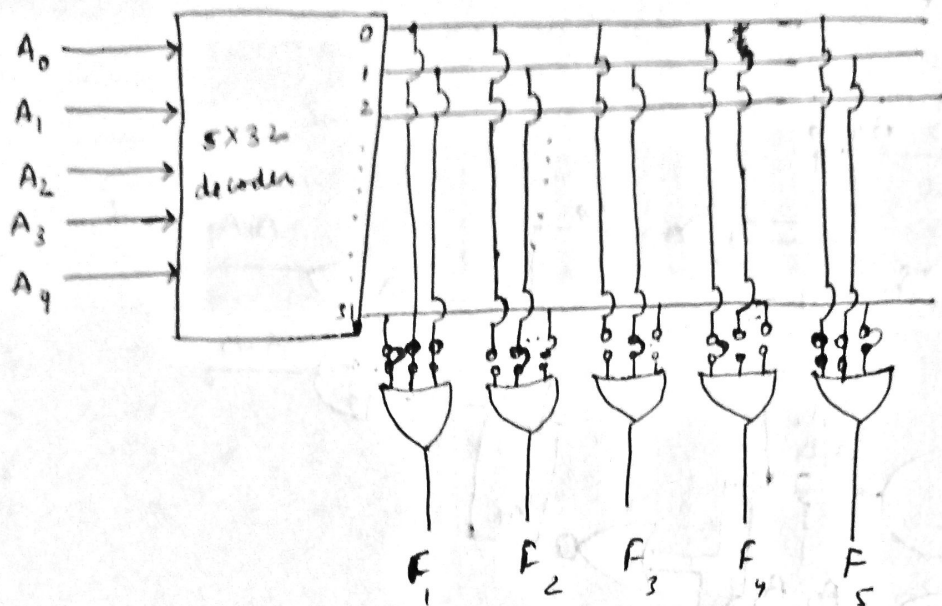
Programmable Logic Devices

It is used to depict different logic gates.



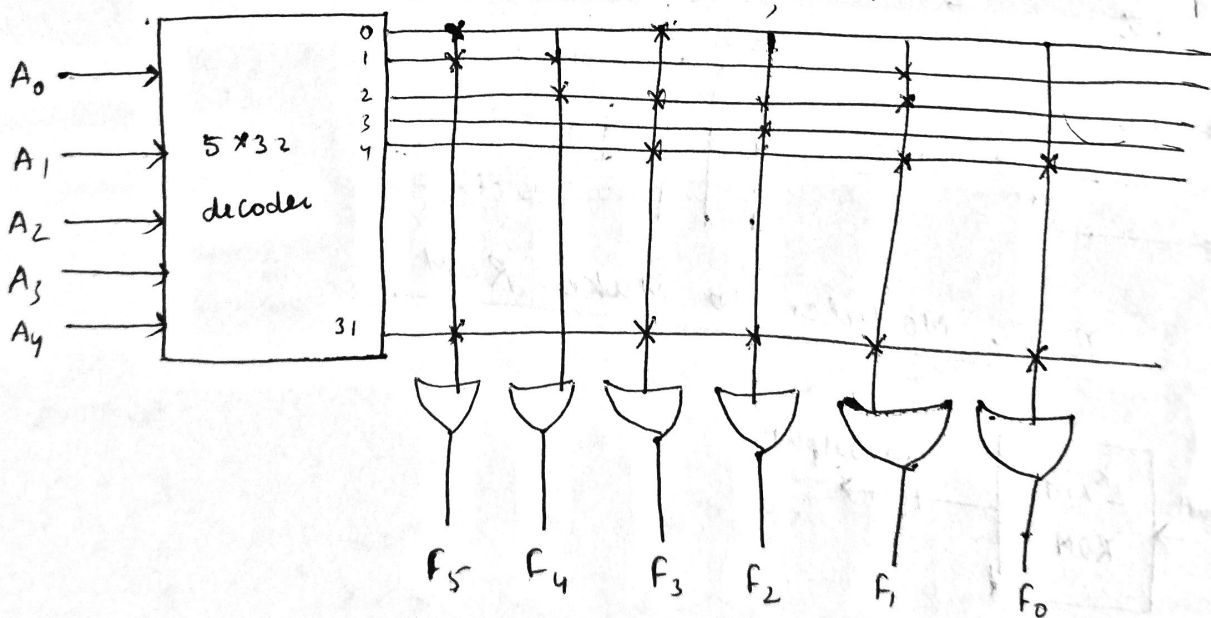
Basic Modules to Make ROM





// functions

A_4	A_3	A_2	A_1	A_0	F_5	F_4	F_3	F_2	F_1	F_0
0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	1	1	1	0	1	0	0
0	0	0	1	0	0	1	1	1	0	0
0	0	0	1	1	0	0	1	0	1	1
...	...	...	...	...	...	...	...	...	...	...



$$a \oplus 0 = a$$

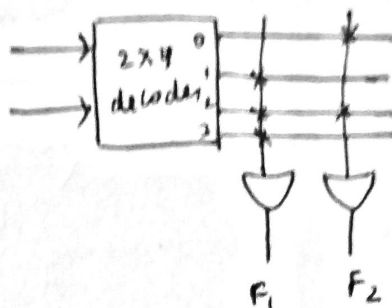
$$a \oplus 1 = \bar{a} + a \cdot 0 = \bar{a}$$

* In ROM, we are taking all minterms.

$$F_1(A, A_0) = \Sigma(1, 2, 3) ; F_2(A, A_0) = \Sigma(0, 2)$$

Gate: Counters
(Imp)

	A_1	A_0	F_1	F_2
m_0	0	0	0	1
m_1	0	1	1	0
m_2	1	0	1	1
m_3	1	1	1	0



27 Feb '23: CE
Monday
Lab (9-11)

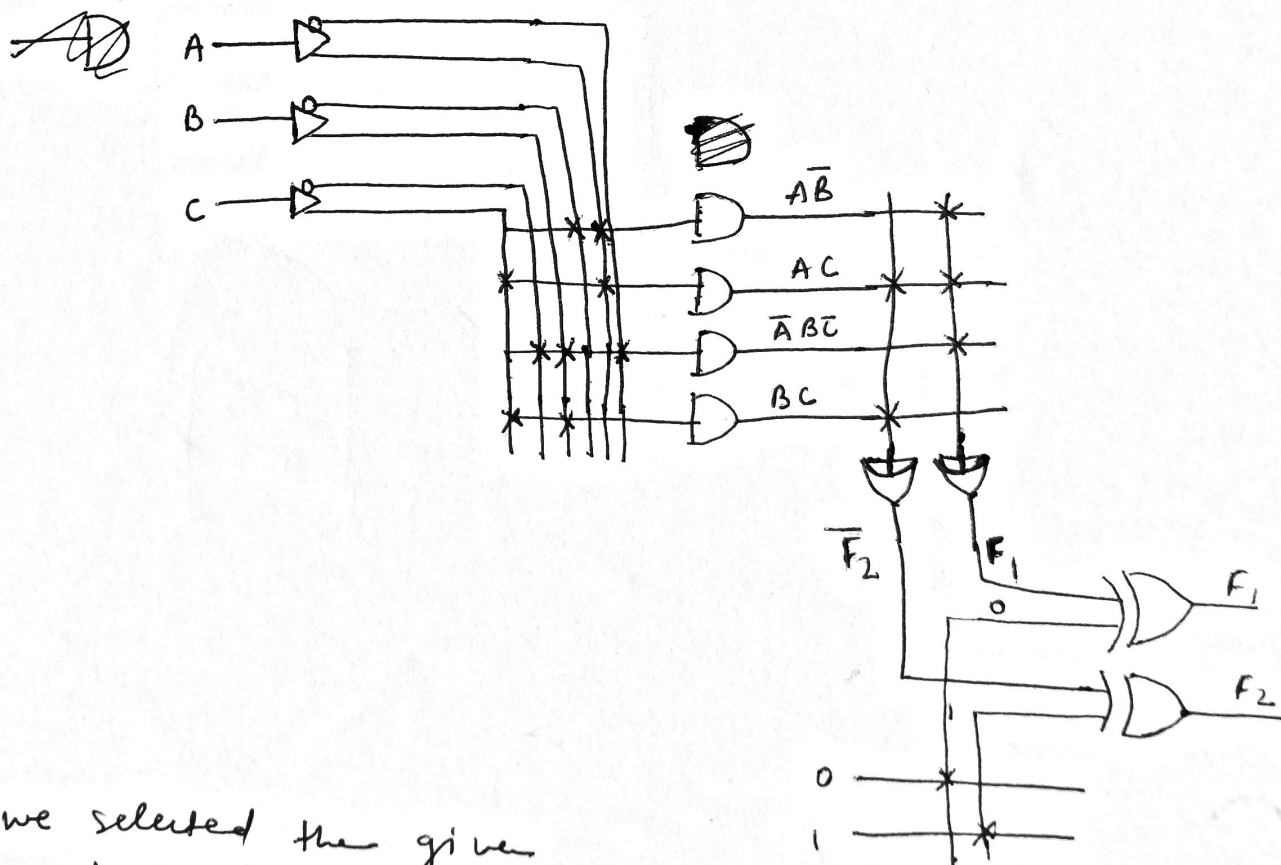
PLA's: Programmable Logic Array

$$F_1 = A\bar{B} + AC + \bar{A}B\bar{C}$$

$$F_2 = \overline{AC + BC}$$

Terms	S.NO.	A	B	C	F_1	$F_2(C)$
$A\bar{B}$		1	0	—	1	—
AC		1	—	1	1	1
$\bar{A}B\bar{C}$		0	1	0	1	—
BC		—	1	1	—	1

Implement



* we selected the given hints in PLA's.